

Transboundary pollution and ecological degradation in the Kura–Araz river system and their implications for water security and green economic resilience

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Keywords

green economy, Kura–Araz, transboundary pollution, water security, river basin management

ABSTRACT

The Kura–Araz river system is central to Azerbaijan's water security, irrigated agriculture, settlements, biodiversity and regional economic resilience. This study integrates two educational research projects that examined river pollution, declining water availability, biodiversity pressure and practical responses from complementary perspectives. The source materials were synthesized with peer-reviewed and institutional evidence and interpreted through water-security and green-economy frameworks. The combined evidence identifies a pressure chain that begins with transboundary and domestic wastewater, industrial and mining emissions, agricultural runoff, solid waste, flow regulation and climate-related water scarcity. Published studies confirm that the Kura–Araz system is exposed to spatially uneven metal pollution and that Azerbaijan depends heavily on transboundary inflows, while recent national water-sector assessments emphasize increasing scarcity and the need for more efficient institutions, monitoring and investment. The original student works also highlight biodiversity loss, degradation of riparian habitats, risks to fisheries and agriculture, and the importance of public awareness, riverbank protection and modern treatment systems. Economically, deteriorating water quality and quantity can increase treatment and monitoring needs, constrain irrigation, reduce ecosystem-service value and weaken rural and urban resilience. The study concludes that basin-scale monitoring, wastewater treatment, efficient irrigation, transboundary cooperation and environmental education should be treated as mutually reinforcing investments in natural capital and Azerbaijan's green economic transition.

INTRODUCTION

The Kura–Araz river system is the principal transboundary freshwater network of the South Caucasus and a strategic natural asset for Azerbaijan. It supports municipal water supply, irrigated agriculture, hydropower, industry, fisheries, riparian habitats and the Caspian Sea receiving environment. This multifunctional role means that river degradation affects not only ecological condition but also food production, public services, human health and regional development. The World Bank (2024) estimates that 78% of Azerbaijan's runoff enters through the transboundary Kura and Araz rivers, underscoring the country's downstream dependence on water quantity and quality formed beyond its borders.

The two source research projects used in this article approach the problem from complementary angles. Muradli's project focused on the contemporary ecological condition of the Kura and Araz, emphasizing transboundary pollution, heavy metals, phenols, petroleum-related pollutants, biodiversity and health risks. The "Kürün səsi" group project examined pollution, declining flow, salinity intrusion, biodiversity loss, dams, agricultural pressure and proposed solutions such as basin cooperation, efficient irrigation, wastewater treatment, rainwater harvesting and environmental awareness. Both projects therefore treat the river as a coupled

ecological and socio-economic system rather than as an isolated watercourse.

The economic dimension is especially important for a green-economy interpretation. Water-quality deterioration can raise treatment and monitoring requirements; reduced flow can intensify competition among irrigation, household and ecosystem needs; biodiversity loss can weaken fisheries and recreation; and salinity or pollution can reduce the productive value of water and land. The World Bank's green-growth analysis for Azerbaijan identifies water and waste management, resource efficiency and climate-smart agriculture among measures capable of improving both environmental performance and economic resilience (World Bank, 2022).

This study asks three questions: (1) which transboundary and domestic pressures on the Kura–Araz system are consistently supported by the student source materials and external evidence; (2) how do these pressures propagate through water quality, hydrology, biodiversity and human use; and (3) which responses can jointly improve water security, ecosystem condition and green economic resilience?

LITERATURE REVIEW

Transboundary contamination in the Kura–Araz basin has long been

documented scientifically. Suleymanov et al. (2010), using repeated sampling from the Azerbaijan part of the basin, found substantial spatial differences in trace-metal concentrations and reported that sites near the borders with Armenia and Georgia were among the most polluted. The study also showed that a large share of several metals was transported with suspended matter, demonstrating that water quality cannot be understood only through dissolved concentrations.

More recent evidence adds a downstream water-quality perspective. Aliyev and Abduev (2025) analyzed monitoring data from 14 points along the Kura for 2020–2022 and found marked changes in major-ion composition and mineralization along the river. Their analysis also identifies industrial, municipal and untreated wastewater discharges as continuing pressures. These findings support the pollution pathways identified in the student projects, while also showing why individual source claims about exact exceedance multiples require verification against monitoring datasets.

Integrated basin management has become a central institutional response. The EU Water Initiative Plus river-basin work for the Kura upstream of Mingachevir applied Water Framework Directive principles to water-body delineation, pressure assessment, monitoring and programme-of-measures planning (EUWI+, 2021). The related River Basin Management Plan included an economic analysis of water use and of proposed measures, explicitly linking ecological objectives to cost recovery, investment and water-service planning.

Azerbaijan's broader water-security context strengthens this linkage. The World Bank (2024) identifies high dependence on transboundary inflows, climate-related scarcity, infrastructure inefficiencies and the need for improved water accounting, wastewater performance and institutional coordination. The Country Climate and Development Report likewise stresses that water scarcity and climate pressures can affect agriculture, livelihoods and economic diversification, making resilient water management part of long-term development strategy (World Bank Group, 2023).

The literature therefore suggests a combined framework in which pollution control, hydrological resilience, ecosystem protection and economic efficiency are mutually dependent. For a downstream country, prevention and cooperation upstream can reduce treatment and restoration burdens downstream, while domestic investments in wastewater treatment, efficient irrigation and monitoring protect the productive value of natural capital.

MATERIALS AND METHODS

The study used an integrative qualitative synthesis of two author-prepared educational research dossiers completed in 2026. The first source was a 13-page presentation by Nihat Muradli on pollution and environmental effects in the Kura and Araz basins. It drew on monitoring reports, scientific literature, laboratory information and field observations and organized evidence around pollutants, ecosystem effects, human health and management responses. The second source was a 15-page project prepared by Asya B., Mahira A. and Sadaqat M. under the “Kürün səsi” group. It examined transboundary pollution, declining river flow, biodiversity, salinity intrusion, dams, agricultural water use, community awareness and proposed interventions.

Because the educational source documents contain several graphs and percentages without an attached raw dataset or complete calculation notes, those figures were not treated as independently verified statistical evidence. Instead, the study retained their qualitative direction and research questions and checked the main mechanisms

against external sources. Exact quantitative claims were used only where they could be supported by peer-reviewed or institutional evidence. External verification relied primarily on Suleymanov et al. (2010), EUWI+ (2021), the World Bank (2022, 2024), the World Bank Group (2023) and Aliyev and Abduev (2025).

The synthesis classified evidence into five pressure categories: transboundary industrial and mining pollution; domestic wastewater and solid waste; agricultural runoff; declining flow and water scarcity; and hydromorphological change from reservoirs and dams. For each category, ecological consequences and plausible economic channels were identified. The analysis is interpretive rather than a monetary valuation, and it does not estimate causal effect sizes or aggregate economic losses.

RESULTS AND DISCUSSION

Transboundary and domestic pollution as a coupled basin pressure

Both educational projects identify transboundary pollution as a defining characteristic of the Kura–Araz system. This framing is consistent with the basin geography: upstream urban, industrial, mining and agricultural activities can affect water entering Azerbaijan. Suleymanov et al. (2010) provides strong empirical support for this mechanism, documenting elevated trace-metal pollution at border-influenced sites and showing that the pollutant burden varies sharply across the basin. Muradli's source project additionally emphasizes phenols, petroleum products and untreated wastewater, while the “Kürün səsi” project highlights industrial discharge, sewage and agricultural chemicals.

Domestic pressures within Azerbaijan remain important as well. Aliyev and Abduev (2025) identify untreated or insufficiently treated municipal and industrial wastewater as continuing contributors to Kura water-quality degradation. This matters because transboundary and domestic pollution interact: imported pollution can enter a river already subject to local wastewater, runoff and solid-waste pressures, while lower flows can reduce dilution capacity. The policy implication is therefore not to treat upstream and domestic sources as competing explanations, but as cumulative pressures requiring coordinated management.

Table 1 summarizes the main pressure pathways retained from the two educational studies and shows how they connect to ecological condition and green-economy resilience.

Water scarcity, flow alteration and biodiversity consequences

The second major theme is the interaction between hydrological decline and pollution. The World Bank (2024) reports that water availability in Azerbaijan is declining under rising upstream demand and climate change, while national dependence on transboundary inflows remains high. Lower river flow can intensify ecological stress by reducing dilution, changing habitat availability and increasing competition among municipal, agricultural and environmental water uses. Both student projects recognize this compound risk and connect it to hotter and drier conditions, water abstraction and reservoir regulation.

Biodiversity impacts follow through several pathways. The source projects discuss degradation of riparian vegetation, pressure on sturgeon and other fish, disruption of migration by dams, eutrophication and loss of habitat quality. While the source documents do not provide a standardized biological monitoring dataset, these mechanisms are consistent with the broader basin-management literature, which treats ecological status and hydromorphology as core dimensions of river health (EUWI+, 2021).

The economic relevance is direct because fisheries, erosion control, water purification, recreation and landscape quality are ecosystem services with social and productive value.

Table 1. Main pressure pathways and their implications for the Kura–Araz river system

Pressure / source evidence	Environmental, water-security and green-economy implication
Transboundary industrial and mining pollution; trace metals documented at border-influenced sites	Toxic stress, sediment contamination and reduced aquatic habitat quality; greater monitoring and treatment burden; cross-border externalities require coordinated prevention.
Municipal wastewater, industrial effluent and solid waste	Organic loading, oxygen stress, litter and degraded riverbank condition; higher treatment and cleanup needs and lower amenity and ecosystem-service value.
Agricultural fertilizers, pesticides and inefficient irrigation	Nutrient loading, eutrophication risk and reduced environmental flows; lower water productivity and weaker farm resilience.
Climate-related scarcity, upstream abstractions and reservoir regulation	Lower dilution capacity, altered habitat and stronger competition among water users; greater need for water accounting, efficiency and adaptive allocation.
Dams and hydromorphological modification	Interrupted fish migration and altered flow regime; trade-offs among energy, irrigation and ecosystem services requiring basin-level optimization.

Water security and green economic resilience

River degradation creates economic exposure through water treatment, agriculture, infrastructure and natural-capital loss. Poorer raw-water quality can require more intensive monitoring and treatment before use. Water scarcity can reduce irrigation reliability, while salinity and pollution can lower the suitability of water for crops and soils. In rural economies, these effects can interact with already variable climate conditions and raise the importance of efficient irrigation, leakage reduction and water accounting. Agriculture is especially sensitive because irrigated production is a major water user in Azerbaijan and an important source of employment and non-oil economic activity (World Bank, 2024).

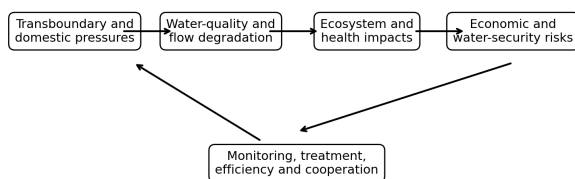


Figure 1. Integrated pathway from river-basin pressures to green-economy responses

The framework in Figure 1 shows that environmental and economic effects are not separate endpoints. Pollution and hydrological pressure first change water quality and flow, then affect ecosystems, health and productive uses, and finally generate costs or lost benefits through treatment, irrigation constraints, degraded fisheries and reduced ecosystem services. Green-economy responses work in the opposite direction by preventing pollution, increasing resource efficiency and protecting natural capital.

Priority responses from treatment and monitoring to basin cooperation

The combined evidence supports a layered response. First, wastewater collection and treatment should be improved in urban and industrial centers because prevention is generally more resource-efficient than repeated downstream cleanup. Second, agricultural measures should reduce fertilizer runoff and improve irrigation efficiency. Third,

monitoring should combine chemical, biological and hydrological indicators and use basin-scale information systems that can detect changing risks. EUWI+ (2021) and the World Bank (2024) both emphasize integrated planning, monitoring and stronger institutional capacity as foundations for more resilient water management.

Transboundary governance, green skills and study limitations

Transboundary cooperation is indispensable because downstream water security cannot be achieved solely through domestic measures. Shared monitoring protocols, data exchange, pollution-event notification and coordinated basin planning can reduce uncertainty and improve investment decisions. The educational projects also place unusual emphasis on public awareness and “green skills”, including school-based water education, riverbank cleanups and responsible household behavior. These actions are not substitutes for infrastructure or regulation, but they can support compliance, community monitoring and long-term stewardship.

The study has several limitations. The two source projects are educational research products rather than standardized basin-monitoring studies, and some of their charts and percentage claims lack accessible raw data. For that reason, the present article uses them primarily to preserve the students’ problem framing, observations and proposed solutions, while quantitative interpretation relies on independently traceable literature. No monetary valuation, laboratory resampling or causal modeling was performed. Future work should integrate standardized water-quality data with economic valuation of treatment costs, irrigation losses and ecosystem-service benefits.

CONCLUSIONS

This study integrates two student-led research projects to examine how transboundary pollution, domestic wastewater, agricultural runoff, water scarcity and hydromorphological change interact in the Kura–Araz river system. External evidence confirms that the basin is exposed to spatially uneven metal pollution, strong dependence on transboundary inflows and increasing water-security pressure. The combined analysis shows that ecological degradation has economic consequences through higher treatment and monitoring needs, irrigation risk, biodiversity loss and weakened ecosystem services. From a green-economy perspective, the most effective response is a portfolio rather than a single intervention: stronger wastewater treatment, efficient irrigation, basin-scale monitoring, water accounting, transboundary cooperation, riverbank and habitat protection, and sustained environmental education. These measures protect water as natural capital while reducing avoidable downstream costs and supporting agriculture, public health and resilient local development. The original educational projects provide a valuable foundation for awareness and problem identification, but future research should pair such engagement with standardized monitoring datasets and explicit economic assessment.

AUTHOR CONTRIBUTIONS

N.M. contributed the original Kura–Araz pollution research, source collection and preliminary environmental synthesis. A.B., M.A. and S.M. contributed the original Kura ecological-problems project, visual synthesis, proposed solutions and survey design. T.M. contributed supervision, methodology development, academic structuring, literature integration and writing—review and editing. All authors reviewed and approved the final manuscript.

CONFLICTS OF INTEREST

The authors declare no conflict of interest.

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This research received no external funding.

DATA AVAILABILITY

This study synthesizes two author-prepared educational research dossiers and publicly available secondary sources. No new laboratory sampling was undertaken. Several source-project graphics were not accompanied by raw datasets; therefore, their exact percentages were not used as independently verified quantitative evidence. The cited institutional and peer-reviewed sources are publicly accessible through the referenced publishers and organizations.

REFERENCES

- Aliyev, E., & Abduyev, M. (2025). Quality trend of Kura River in Azerbaijan by major ion concentration. *Discover Water*, 5, 82. <https://doi.org/10.1007/s43832-025-00219-9>
- European Union Water Initiative Plus (EUWI+). (2021). Draft river basin management plan for the Kura upstream of Mingachevir Dam river basin district. European Union Water Initiative Plus.
- Ministry of Ecology and Natural Resources of the Republic of Azerbaijan. (2024). Monitoring report on transboundary rivers of Azerbaijan. Baku, Azerbaijan.
- Suleymanov, B., Ahmedov, M., Safarova, K., & Steinnes, E. (2010). Metals in main rivers of Azerbaijan: Influence of transboundary pollution. *Water, Air, & Soil Pollution*, 213(1–4), 301–310. <https://doi.org/10.1007/s11270-010-0385-1>
- United Nations Economic Commission for Europe. (2011). Second assessment of transboundary rivers, lakes and groundwaters. United Nations.
- World Bank. (2022). Azerbaijan: Towards green growth—Issues note. World Bank.
- World Bank. (2024). Azerbaijan: Transitioning to efficient water sector institutions and programs for addressing water security challenges. World Bank.
- World Bank Group. (2023). Azerbaijan country climate and development report. World Bank. <https://doi.org/10.1596/40622>
- Muradli, N. (2026). Çay suyunun çirklənməsi və onun ətraf mühitə təsiri [Educational research project]. Aghali School, Zangilan, Azerbaijan.
- Asya, B., Mahira, A., & Sadaqat, M. (2026). Kür çayının ekoloji problemləri və onların həlləri [Educational research project]. Kürün səsi group, Baku, Azerbaijan.